

HARIHARAN NARLAKANTI

AI Engineer | Backend Engineer | Software Engineer

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SUMMARY

Computer Science graduate (AI & ML) with 8.89/10 CGPA focused on AI and backend engineering. Built and deployed AI applications using Python, FastAPI, Hugging Face Transformers, RAG, semantic search, Docker, and GitHub Actions. Skilled in REST API development, document intelligence, embedding-based retrieval, NLP model evaluation, testing, and production-oriented deployment.

EDUCATION

Malla Reddy University, Hyderabad

2022 – 2026

B.Tech in Computer Science & Engineering (AI & ML)

CGPA: 8.89/10

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript

Backend & APIs: FastAPI, REST APIs, API Design, Pydantic, JWT Authentication, Node.js, Express.js

AI/ML: NLP, Hugging Face Transformers, PyTorch, SentenceTransformers, RAG, Embeddings, QLoRA, PEFT

Databases: PostgreSQL, MySQL, MongoDB, ChromaDB

Testing & DevOps: pytest, Git, GitHub Actions, CI/CD, Docker, Linux

Deployment: Hugging Face Spaces, Vercel

EXPERIENCE

Virtual Internship – Full Stack Developer - MERN Stack, SmartInternz

19 May 2025 – 05 July 2025

- Built **SocialeX**, a MERN social networking application using React, Node.js, Express.js, and MongoDB with authenticated REST APIs and JWT-based access control.
- Implemented real-time one-to-one messaging with **Socket.IO** and enforced server-side authorization across user profiles, posts, and protected resources.

PROJECTS

Saaram – AI-Powered Telugu News Summarization & Speech Platform

GitHub | Live Demo

- Built and deployed a full-stack Telugu news summarization and speech platform using **Python, FastAPI, React/Vite, Hugging Face Transformers, and Docker**.
- Developed a hybrid NLP pipeline combining morphology-aware **TF-IDF**, pretrained **mT5**, and parameter-efficient **QLoRA** fine-tuning for resource-constrained inference.
- Evaluated **5 configurations on 1,302 XL-Sum Telugu samples**; fine-tuned mT5 achieved **0.1639 ROUGE-1, 0.0496 ROUGE-2, and 0.1452 ROUGE-L**.
- Improved character n-gram TF-IDF **ROUGE-2 by 26.6%** over the word-level baseline and implemented adaptive routing with resource-aware TF-IDF fallback.
- Automated backend deployment to Hugging Face Spaces using **Docker, GitHub Actions, and OIDC-based Trusted Publishers**.

ResearchCompass – AI Research Paper Analyzer

GitHub | Live Demo

- Built a document intelligence platform using **Next.js, FastAPI, PyMuPDF, SentenceTransformers, ChromaDB, and Groq** for academic paper analysis and semantic retrieval.
- Designed modular REST API workflows for PDF ingestion, section-aware chunking, embedding generation, vector retrieval, and structured LLM inference using **Pydantic**.
- Implemented targeted multi-query **RAG**, cross-paper comparison, semantic search, and evidence-grounded literature review generation.
- Developed **53 unit and integration tests with pytest** and configured automated backend deployment using **GitHub Actions** and Hugging Face Spaces.

Smart-FAQ – Semantic Search Platform

GitHub

- Developed a full-stack semantic FAQ platform using **FastAPI, React/Vite, MongoDB, and SentenceTransformers** for natural-language search.
- Implemented embedding-based semantic retrieval, confidence scoring, fallback handling, and JWT-protected administrative REST APIs.

ACHIEVEMENTS & RESEARCH

- Research Presenter, CIS 2026** – Presented Saaram research at the 7th Congress on Intelligent Systems, NIT Warangal.
- Best Project Award**, Malla Reddy University – Recognized for the Telugu NLP news summarization project.

CERTIFICATIONS

- Machine Learning Specialization** – DeepLearning.AI & Stanford University, 2026
- IBM Full Stack Software Developer Professional Certificate** – IBM / Coursera, 2024